Chunking and Overlap in RAG

What is chunking?

Chunking is the process of dividing a long document into smaller text passages before creating embeddings. A retrieval system usually searches these smaller passages instead of comparing a user question with an entire document. Smaller chunks can make retrieval more precise because a chunk can focus on a narrower topic.

Why use overlap?

Chunk boundaries can split an important sentence or idea between two chunks. To reduce this problem, neighboring chunks often share a small amount of overlapping text. For example, if a chunk contains 120 words, the next chunk might repeat the final 20 words of the previous chunk. This helps preserve context near the boundary.

Choosing chunk size

Very small chunks may lose useful context, while very large chunks may contain several unrelated ideas and reduce retrieval precision. The best chunk size depends on the document type, question style, and embedding model. In a small educational RAG project, a simple fixed-size chunking strategy is usually sufficient for initial experiments.

Indexing behavior

Chunking is normally performed when a document is ingested into the knowledge base. Each chunk receives its own embedding and metadata such as the source file name, page number, and chunk number. If the document has not changed, the stored chunks and embeddings do not need to be rebuilt for every user question.